

Answer Key — Biology Grade 11

Sub-Strand 3.1: Reproduction in Animals • Lesson 5: From Fertilisation to Birth: How Hormones and the Placenta Sustain Pregnancy

Teacher reference only. Do not project or hand out.

Quick marking grid

Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9
D	A	A	A	A	D	B	A	B

USE AFTER: PREDICT PHASE — Predict: sketch of how a baby inside a mother gets food and oxygen

Q1. A cow carries one calf for about 9 months. How does the developing calf get its oxygen and food while inside the mother?

- A. The calf swallows food through its mouth and breathes air through its nose inside the uterus
- B. The mother's blood flows directly into the calf's blood vessels and mixes with it
- C. The calf uses only the yolk stored inside its body until birth
- D. Substances diffuse across the thin membranes of the placenta between the mother's blood and the calf's blood**

Correct answer: D. In mammals, the placenta lets oxygen and nutrients diffuse from mother to fetus across thin membranes, with no mixing of blood. Feeding by mouth, direct blood mixing and a yolk-only supply are the common misconceptions.

USE AFTER: PREDICT PHASE — Predict: recording misconceptions such as 'IVF babies grow in test tubes'

Q2. Which statement about IVF babies is correct?

- A. The embryo is placed in the mother's uterus and still develops inside her body**
- B. The baby grows wholly in a test tube until birth
- C. The baby grows inside the mother only for the first week and then in a machine
- D. The baby does not need a placenta because it started in a laboratory

Correct answer: A. The lesson corrects the misconception that IVF babies grow in test tubes: embryos are placed in the uterus and develop with a placenta like any other pregnancy.

USE AFTER: OBSERVE PHASE — Observe: animation clips 1 and 2 (fertilisation in the oviduct, implantation and early hCG signalling)

Q3. In humans, where does fertilisation normally take place, and where does the embryo then implant?

- A. Fertilisation in the oviduct; implantation in the uterine wall**
- B. Fertilisation in the vagina; implantation in the cervix
- C. Fertilisation in the uterus; implantation in the oviduct
- D. Fertilisation in the ovary; implantation in the placenta

Correct answer: A. The egg and sperm meet in the oviduct, and the embryo then embeds in the lining of the uterus (implantation). The placenta forms afterwards from this implantation.

USE AFTER: OBSERVE PHASE — Observe: drawing and labelling the placenta and umbilical cord, showing where exchange occurs

Q4. A learner draws a placenta and marks where oxygen and nutrients pass from the mother to the baby. Which labelled part is the site where this exchange happens?

- A. The placenta, across thin membranes**
- B. The wall of the vagina
- C. The amniotic fluid, where the baby absorbs food
- D. The umbilical cord, where maternal blood mixes with fetal blood

Correct answer: A. Exchange occurs at the placenta by diffusion across thin membranes. The umbilical cord only carries fetal blood to and from the placenta, and the amniotic fluid protects the baby.

USE AFTER: EXPLAIN PHASE — Explain: sentence frames 'The placenta allows ___ because ___' and addressing the misconception that blood mixes

Q5. Why does the mother's blood not mix with the baby's blood in the placenta, even though substances pass between them?

- A. They have separate blood supplies, and substances move by diffusion across thin membranes**
- B. The umbilical cord pumps the mother's blood into the baby
- C. The mother's red blood cells are too small to enter the baby's blood
- D. Blood mixes only during labour, not during pregnancy

Correct answer: A. Maternal and fetal blood stay in separate vessels. Oxygen, nutrients and wastes diffuse across the thin placental membranes, so the blood itself never mixes.

USE AFTER: EXPLAIN PHASE — Explain: building the hormonal flowchart (hCG keeps the corpus luteum active, progesterone maintains the lining)

Q6. After implantation, which hormone from the embryo keeps the corpus luteum active so that progesterone continues to maintain the uterine lining?

- A. Oestrogen in late pregnancy
- B. Prolactin
- C. Oxytocin
- D. hCG**

Correct answer: D. hCG signals the corpus luteum to stay active, so it keeps producing progesterone. Later the placenta takes over hormone production.

USE AFTER: EXPLAIN PHASE — Explain: flowchart showing oxytocin triggering contractions (positive feedback)

Q7. At the end of pregnancy, oxytocin causes the uterus to contract. These contractions push the baby against the cervix, which causes more oxytocin to be released. What is this type of control called?

- A. Hormone mixing
- B. Positive feedback**
- C. Negative feedback
- D. Homeostasis

Correct answer: B. Each contraction leads to more oxytocin and stronger contractions, so the effect increases the cause. That is positive feedback, and it continues until birth.

USE AFTER: MODEL BUILDING PHASE — Model: comparison table of frog, hen and human/viviparous mammal (number of offspring, fertilisation type, nutrition source, parental care)

Q8. A frog releases thousands of eggs in a pond, while a cow produces one calf at a time. Which statement best explains this difference?

- A. The cow's calf is protected and nourished inside the mother and cared for after birth, so fewer offspring are needed**
- B. Frog eggs have thick shells, so many are needed to break through them
- C. The frog's eggs are fertilised inside the body, so many are needed
- D. The cow has fewer eggs because cows are larger than frogs

Correct answer: A. Frogs use external fertilisation with little parental care, so many eggs are lost and many are needed. The cow's high parental investment means one well-supported calf is likely to survive.

USE AFTER: END OF LESSON — Model: exit ticket (one hormone and its role) and revised class model

Q9. Which pair correctly matches a hormone with its role in pregnancy, birth or lactation?

- A. Oxytocin: stops the uterine lining from forming
- B. Prolactin: supports milk production after birth**
- C. Progesterone: causes labour contractions
- D. hCG: produced by the mother's breasts to make milk

Correct answer: B. Prolactin supports milk production. Oxytocin triggers contractions, progesterone maintains the uterine lining, and hCG signals the corpus luteum to remain active.

